Patient Shortage in Histopathology: Breadth, Depth, and Effective Support

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Abstract

Training on fewer patients impairs histopathology patch classification even when class patch counts are held fixed, and neither a different downstream classifier nor an equalized training objective recovers the loss. This experiment separates patient breadth from patch depth on a three-by-three allocation grid in BRACS and TCGA-UT, using frozen Virchow2 features and fixed patient-disjoint splits. All cells draw from common patient pools, and depth comparisons at fixed breadth reuse the same patients with nested patches. At 160 patches per class, allocating eight patches to each of twenty patients instead of 32 patches to each of five patients improves patient-macro balanced accuracy by 3.71 percentage points on BRACS (95% interval: 2.21 to 5.35) and 9.52 points on TCGA-UT (8.96 to 10.07), and lowers macro negative log-likelihood even after temperature scaling. On BRACS, redundancy-adjusted effective support describes the grid closely and absorbs nearly all of the breadth effect. On TCGA-UT, whose patches are repeated crops of a few pathologist-selected uniform tumour regions per slide, additional depth adds less than half a point, and patients contribute more than their effective observations. Within the tested range, spreading a fixed labelling budget over more patients is thus a budget-neutral way to train more accurate classifiers with more informative probabilities than deeper annotation of fewer patients.

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1 Motivation

The signal taxonomy of Queisler (2026c) separates *nominal support*, the number of labelled patches n_c available for class c, from *independent support*, the number of distinct patients G_c supplying them. Patches from one patient share acquisition conditions and much of their morphology, so a large patch count assembled from few patients need not carry the evidence its size suggests. The same taxonomy already anticipates a quantitative version of this argument. Its *redundancy-adjusted effective support* $N_{\mathrm{eff},c} = n_c/\mathrm{DE}_c$ divides the patch count by a design effect $\mathrm{DE}_c = 1 + (m_c^\star - 1)\,\mathrm{ICC}_c$, where $m_c^\star$ is the size-biased number of patches per patient and ICC_c is the within-patient correlation of a scalar summary of the representation (Kish, 1965; Rutterford et al., 2015).

Three earlier results define what is left to explain. The controlled shortage benchmark reduced patient support while holding class patch counts fixed and measured discrimination losses of 2.62–3.03 percentage points on BRACS and 4.76–4.83 points on TCGA-UT, which the tested mitigation methods recovered by between −6.5% and 2.5% (Queisler, 2026b). The classifier comparison then showed that the deficit is not a property of the downstream model: replacing the multilayer perceptron by multinomial logistic regression or cosine nearest neighbours improved concentrated-support accuracy by at most 0.48 points, while broader coverage benefited every classifier (Queisler, 2026e). The influence comparison finally showed that the deficit is not an artefact of unequal loss contributions: patient-average training changed accuracy by −1.36 points on BRACS and −0.004 points on TCGA-UT, leaving residual coverage benefits of 2.73 and 4.70 points (Queisler, 2026d). The bottleneck has thus been localized to the training material itself, but its content has not been characterized.

Two readings of that material remain. Under *within-patient redundancy*, patches from one patient are statistically dependent, so a patient's contribution is worth fewer independent observations than its patch count states; the deficit is then a shortage of effective observations and should be summarized by a single scalar. Under *coverage beyond redundancy*, patients differ in ways that no scalar correlation captures, because additional patients introduce morphological or acquisition modes absent from the observed ones; the deficit is then a shortage of distinct conditions rather than of effective observations. These readings are not mutually exclusive, and both are compatible with every result reported so far, because the previous allocations varied patient count and within-patient depth together.

Separating them requires an allocation in which the number of contributing patients and the number of patches taken from each of them move independently. This experiment therefore constructs a *breadth* × *depth* grid, with breadth G the number of patients contributing to a class and depth m the number of patches drawn from each of them, and compares allocations that differ in composition at an identical patch budget. Holding depth equal across patients has a second benefit: it removes the contribution inequality studied by Queisler (2026d) by construction, so patch-average and patient-average training coincide and the two objectives need not be compared again.

2 Study design

BRACS retains its seven lesion classes and TCGA-UT its 30 eligible cancer types. Both keep the benchmark dataset eligibility rules, class order, and frozen 2560-dimensional Virchow2 features of the benchmark, formed by concatenating the CLS token and the mean patch token (Zimmermann et al., 2024; Queisler, 2026a). The three existing patient-disjoint train–validation–test splits remain fixed, and BRACS patients and TCGA-UT participants are called patients throughout. Validation and test partitions keep their natural distributions and are never resampled.

Each training allocation is described by one pair (G, m) applied to every class: exactly G patients contribute to class c, and each contributes exactly m patches of that class, so $n_c = Gm$

for all classes. Class patch counts are therefore equal by construction, which removes prevalence as a competing signal and leaves support as the only quantity that varies. Table 1 lists the nine cells of the grid and their per-class patch budgets. Breadth ranges over $G \in \{5, 10, 20\}$ and depth over $m \in \{8, 16, 32\}$, a fourfold range in each factor and a sixteenfold range in n_c. The largest cell uses 4,480 training patches on BRACS and 19,200 on TCGA-UT.

	Patches per patient m		
Patients per class G	8	16	32
5	40	80	160
10	80	160	320
20	160	320	640

Table 1: Per-class patch budget $n_c = Gm$ for the nine allocation cells. Cells sharing a budget form equal-budget sets: $n_c = 160$ contains $(20, 8)$, $(10, 16)$, and $(5, 32)$, while $n_c = 80$ and $n_c = 320$ contain two cells each. Comparisons within such a set differ only in how a fixed number of patches is distributed over patients.

The upper end of the grid is set by the rarest class rather than by cost. Breadth 20 requires twenty patients with at least 32 patches of every class in every training split, and on BRACS the smallest such pool contains only 26 patients, so the next breadth level of 40 is not attainable. Even the largest budget remains below the benchmark's balanced allocations of about 2,400 patches per class on BRACS and about 1,494 on TCGA-UT (Queisler, 2026b). Absolute accuracies are therefore not directly comparable with those reports; the object of interest is the shape of the support surface.

2.1 Allocation rules

Three rules turn a grid cell (G, m) into a concrete training set. Each rule ensures that moving along one axis of the grid changes only that axis.

The first rule gives all cells one patient pool. A patient can contribute m patches of class c only if that many exist. If every depth used its own pool, deeper cells would draw from patients with more tissue, and a depth contrast would partly compare different kinds of patients. All nine cells therefore draw from the same pool, the training patients with at least 32 patches of the class. With N_{irc} the number of patches of class c from patient i in the training partition of split r, the pool is

$$\mathcal{E}_{rc} = \{i : N_{irc} \geq 32\}. \tag{1}$$

Before any model was fitted, every class in every split was confirmed to contain at least 20 eligible patients, the largest breadth in the grid.

The second rule keeps the same patients at every depth. For each split, class, breadth G, and repeated draw d, G patients are drawn from $\mathcal{E}_{rc}$ without replacement, using a random seed that does not depend on depth. The three cells of one breadth and draw therefore contain the same patients. Classes are sampled independently, so a patient eligible for several classes may contribute to each of them.

The third rule nests the patches across depths. The patches of each selected patient are ordered once, cycling through the patient's slides in a fixed order, and a cell of depth m takes the first m patches of that list. The patches of a shallow cell are thus contained in those of every deeper cell,

$$\mathcal{P}_{ircd,8} \subset \mathcal{P}_{ircd,16} \subset \mathcal{P}_{ircd,32}. \tag{2}$$

A depth contrast at fixed breadth therefore adds patches to an unchanged set of patients instead of exchanging one sample for another. Each cell is drawn $R = 5$ times, draw indices are paired across depths, and every cell quantity averages over the draws. Between-draw dispersion is reported separately as a descriptive measure of sensitivity to patient composition.

The readout is the deterministic multinomial logistic regression of Queisler (2026e), with an unpenalized intercept and the same feature preprocessing and convergence criteria. Its regularization strength is selected on the validation partition separately for every split, cell, and draw from the inherited grid, because the training-set size varies by a factor of sixteen across the grid and a single inherited value would confound support with regularization. Only converged fits enter the analysis. The multilayer perceptron and the nearest-neighbour readout are omitted: Queisler (2026e) established that the coverage effect is present under all three, and a single deterministic readout keeps the grid small enough to be run without repeated initializations.

3 Effective support

Equal depth simplifies the redundancy adjustment considerably. With every contributing patient supplying m patches, the size-biased patient size $m_c^\star = \sum_g m_{gc}^2/n_c$ equals m exactly, so the design effect and the effective support of Queisler (2026c) become

$$\mathrm{DE}_c(m) = 1+(m-1)\,\mathrm{ICC}_c, \qquad N_{\mathrm{eff},c}(G,m) = \frac{Gm}{1+(m-1)\,\mathrm{ICC}_c}, \qquad \lim_{m\to\infty} N_{\mathrm{eff},c} = \frac{G}{\mathrm{ICC}_c}. \quad (3)$$

The two grid factors therefore enter effective support asymmetrically. Breadth scales it proportionally, whereas depth scales it only until the design effect grows with m, after which additional patches from the same patients contribute progressively less. The limit makes the prediction concrete: under the redundancy account no depth can raise effective support beyond $G/\,\mathrm{ICC}_c$, and the transition occurs near $m \approx 1/\,\mathrm{ICC}_c$. Whether that transition is visible within the depth ladder is an empirical question about the representation, and one that the estimated correlations settle before any classifier is fitted.

3.1 Estimating the intraclass correlation

The correlation is estimated per class and split from the training partition only, following the procedure fixed in the taxonomy. Each frozen Virchow2 embedding is projected onto the leading principal direction of a principal component analysis fitted jointly across classes on a class-balanced reference sample, which yields a scalar score that does not depend on any fitted classifier. The one-way random-effects estimator with the unequal-cluster-size correction is then applied to these scores with patients as clusters (Kenny and La Voie, 1985), from a random sample of at most 200 patients and at most 200 patches per patient. Class correlations are averaged over the three training splits before evaluating Equation (3); the cell predictor then averages the resulting $N_{\mathrm{eff},c}$ over classes. Correlations characterize the training partitions rather than only the maximum-depth eligible pools.

The two caps bound the cost of loading features and follow from an explicit precision target: the relative standard error of $\widehat{\rho}$ should not exceed 10% wherever the cohort supplies enough patients. For k patients with n patches each and true correlation ρ, the large-sample variance of the one-way estimator is approximately (Donner, 1986)

$$\mathrm{Var}(\widehat{\rho}) \approx \frac{2(1-\rho)^2\,[1+(n-1)\rho]^2}{n(n-1)(k-1)}, \quad (4)$$

so the relative standard error splits into a patient factor and a patch factor,

$$\frac{\sqrt{\mathrm{Var}(\widehat{\rho})}}{\rho} \approx \underbrace{\sqrt{\frac{2}{k-1}}\,(1-\rho)}_{\text{patients}} \cdot \underbrace{\frac{1+(n-1)\rho}{\rho\sqrt{n(n-1)}}}_{\text{patches}}. \quad (5)$$

The patch factor exceeds one and falls towards one as n grows, so no number of patches can push the error below the patient factor. The patient cap therefore follows first: the patient factor stays below 10% for every correlation only if $k \geq 201$, which the cap of 200 patients meets up to rounding. Given 200 patients, the patch cap follows from the weakest correlation the target should cover. For correlations of at least 0.075, a range that contains every class estimate in Table 9, the target requires at least 165 patches per patient; 200 patches give 9.9%, whereas 32 patches would give 13%. The design effect $\mathrm{DE}_c(32) = 1 + 31\rho_c$ is estimated more precisely, at about 7% under the same conditions, because its constant term carries no estimation error.

The target is met wherever the cohort allows it. The patient cap binds only on TCGA-UT, in 36 of its 90 class–split combinations, and Equation (5) evaluated at each class's median patient size gives relative standard errors between 4% and 16% (median 7%). All seven TCGA-UT classes above 10% have fewer than 90 training participants. BRACS training partitions contain only 30 to 69 patients per class, with relative standard errors between 15% and 25% per split before averaging over splits. Larger caps would not change these values: sampling every training patient with all of their patches gives 3% to 16% on TCGA-UT and 15% to 25% on BRACS. The remaining imprecision is thus a property of the cohorts rather than of the caps, and Section 5.5 examines how strongly the conclusions depend on it.

Reporting these correlations serves a purpose beyond the support surface. Each ICC_c is the share of projected feature variance that lies between patients rather than within them, so the class-level table is a direct measurement of intra- and inter-patient heterogeneity for every lesion type and cancer type. Classes whose appearance varies strongly within a patient should have low correlations and lose comparatively little from concentrated sampling, and comparing these estimates with the class-level accuracy effects is an exploratory part of the analysis.

Equation (3) lets the three accounts of Section 1 be stated as separable predictions about the grid, summarized in Table 2. The predictions are distinguishable because $n = Gm$, G, and N_{eff} are different functions of the two grid factors: an equal-budget set holds n constant while G and N_{eff} still vary across its cells.

Account	Claim about the deficit	Prediction on the grid
Nominal support	Only the amount of labelled material matters	Accuracy is a function of $n = Gm$; equal-budget cells agree
Redundancy-adjusted support	Within-patient patches are correlated, so they are worth fewer independent observations	Accuracy is a function of N_{eff}; depth saturates near $m \approx 1/\,\mathrm{ICC}$; cells collapse onto one curve
Coverage beyond redundancy	Patients supply variation that a scalar correlation cannot express	Breadth remains beneficial at matched N_{eff}; a positive residual breadth term persists

Table 2: The three accounts and the grid behaviour each of them implies. The accounts are ordered by how much structure they attribute to the patient grouping. The first is already disfavoured by the benchmark result and is retained as a reference line within this design.

4 Evaluation

The primary endpoint is *patient-macro balanced accuracy*, defined exactly as in the two preceding experiments so that effect sizes remain comparable. For split r, let $\mathcal{P}_{rc}$ contain the test patients with at least one patch of true class c, and let $\mathcal{J}_{irc}^{\text{test}}$ contain that patient's evaluated patches of that class. The split score and the underlying within-patient recalls are

$$A_r = \frac{1}{K}\sum_{c=1}^{K}\frac{1}{|\mathcal{P}_{rc}|}\sum_{i\in\mathcal{P}_{rc}} q_{irc}, \qquad q_{irc} = \frac{1}{|\mathcal{J}_{irc}^{\text{test}}|}\sum_{j\in\mathcal{J}_{irc}^{\text{test}}} \mathbf{1}\{\widehat{y}_j = c\}, \tag{6}$$

where K is the number of classes. A cell score $A(G, m)$ averages the three split scores and the R draws equally. Secondary endpoints are macro negative log-likelihood in nats and expected calibration error over ten equal-width confidence bins, which connect this grid to the probability-quality damage established by Queisler (2026b), together with patch-micro balanced accuracy and class-specific patient-macro recalls. Both probability-quality endpoints are computed after post-hoc temperature scaling (Guo et al., 2017). For every fit, the logits are divided by one positive temperature that minimizes the negative log-likelihood of the natural validation partition, and the same temperature is applied to the test logits. In the benchmark, this single scalar removed much of the probability deficit caused by nominal class deprivation, but only part of the deficit caused by patient shortage (Queisler, 2026b), so scaled probabilities isolate the differences that a practitioner could not remove at negligible cost. Because a positive temperature preserves the ranking of the classes, the accuracy endpoints are unaffected. Unscaled values are reported alongside for comparison.

Three contrasts summarize the surface, all expressed in percentage points:

$$\Delta_m(G) = A(G, 32) - A(G, 8), \qquad \text{depth gain at fixed breadth,} \tag{7}$$

$$\Delta_G(m) = A(20, m) - A(5, m), \qquad \text{breadth gain at fixed depth,} \tag{8}$$

$$X = A(20, 8) - A(5, 32), \qquad \text{equal-budget breadth advantage.} \tag{9}$$

The equal-budget advantage X compares the broadest and the deepest allocation of the $n_c = 160$ set and is the most informative single quantity in the design, because both of its terms consume the same labelling budget and differ only in how that budget is spread over patients. The intermediate cell $(10, 16)$ completes the set and indicates whether the exchange between breadth and depth is monotone.

4.1 Support surface

The contrasts describe individual comparisons, whereas the accounts of Table 2 concern the grid as a whole. Cell accuracies are therefore also summarized by least-squares fits over the nine cells, using the candidate predictors of Equation (3) on a logarithmic scale,

$$A = \alpha + \beta \log n, \qquad A = \alpha + \beta \log G, \qquad A = \alpha + \beta \log N_{\text{eff}}, \tag{10}$$

which have equal numbers of parameters and can be ranked directly by residual standard deviation. Two augmented fits then separate the accounts. Adding breadth to the nominal predictor, $A = \alpha + \beta \log n + \gamma_n \log G$, asks whether patient count matters at all once the patch budget is known, and so reproduces the benchmark finding within this design. Adding breadth to the effective predictor, $A = \alpha + \beta \log N_{\text{eff}} + \gamma_e \log G$, asks the question this experiment exists for: whether patients still matter once their redundancy has been accounted for. The second fit nests the third model of Equation (10) at $\gamma_e = 0$.

Uncertainty follows the procedure of the two preceding experiments. Each of 2,000 replicates of a paired Bayesian bootstrap assigns random positive weights drawn from a Dirichlet$(1, \ldots, 1)$ distribution to the distinct test patients (Rubin, 1981), with every patient's weight shared across cells, draws, splits, classes, and repeated test appearances. Within-class recalls, cell scores, contrasts, and surface coefficients are recomputed on each replicate, and the 2.5th and 97.5th percentiles form individual exploratory 95% intervals without adjustment for multiple comparisons.

These intervals describe held-out patient variation conditional on the realized training draws; variation from the draws themselves is reported separately as the between-draw dispersion of Section 2.1.

4.2 Interpretation criteria

One percentage point remains the prespecified practical-relevance threshold (Queisler, 2026e). A point estimate above one point is practically promising, a lower interval bound above one point supports an effect exceeding the threshold, and an interval crossing zero leaves the direction open. For the surface coefficients the threshold is expressed on the same scale by noting that doubling breadth changes $\log G$ by $\log 2$, so a coefficient matters practically when $\gamma \log 2$ exceeds one point, that is when γ exceeds 1.44 points per log unit.

The accounts are then read as follows. Support that depends only on the patch budget requires both X and γ_n to be indistinguishable from zero, which the benchmark result makes unlikely and which therefore functions as a check on the design rather than as a live hypothesis. Redundancy-adjusted support is supported when the effective predictor attains the smallest residual standard deviation of the three fits in Equation (10) and the interval for γ_e lies inside the practical band around zero. Coverage beyond redundancy is supported when the interval for γ_e excludes zero with a positive point estimate, and is practically relevant when that estimate also exceeds 1.44 points per log unit. Because both augmented fits are conditional on the estimated correlations, a positive γ_e can also indicate that the scalar projection underestimates within-patient dependence, so the class-level correlation table is consulted before the residual term is read as missing patient variation.

Depth saturation is judged separately from the fits. It is supported when $\Delta_m(20)$ has an interval contained in ± 1 point, that is when quadrupling the patches taken from each of twenty patients leaves accuracy practically unchanged. This criterion is informative only if the predicted transition falls inside the ladder, so the mean estimated correlation is compared with the depth levels in advance: a mean ICC below $1/32$ places the transition beyond the deepest cell, and an absence of saturation is then uninformative about the redundancy account rather than evidence against it.

5 Results

Broader patient sampling improves accuracy and probability quality at a fixed patch budget in both datasets. On BRACS, additional patches from the same patients also help, and effective support describes the grid closely. On TCGA-UT, depth gains are small, breadth dominates, and patients contribute more than their effective observations.

5.1 Accuracy grid

The grid can be read in three directions. Moving down a column adds patients at fixed depth, moving along a row adds patches from the same patients, and the anti-diagonal of 160 patches per class keeps the budget fixed while shifting it from depth to breadth. If patches were interchangeable regardless of the patient they come from, accuracy would depend only on the budget, and the three cells of the anti-diagonal would agree.

Every class in every split contained enough eligible patients for the full grid: 26 to 56 per class and split on BRACS and 30 to 514 on TCGA-UT. All 135 selected fits per dataset converged, covering nine cells, three splits, and five draws, so each cell of Table 3 averages fifteen fitted allocations.

BRACS accuracy rises from 34.55% at $(5, 8)$ to 42.88% at $(20, 32)$; TCGA-UT rises from 61.97% to 72.08%. Point estimates increase with both factors throughout both grids. At 160 patches per class, BRACS accuracy progresses from 37.13% at $(5, 32)$ through 38.69% at $(10, 16)$ to 40.84%

BRACS

G	$m = 8$	$m = 16$	$m = 32$
5	34.55 [32.24, 36.91]	35.73 [33.16, 38.27]	37.13 [34.50, 39.78]
10	37.72 [35.25, 40.22]	38.69 [35.67, 41.70]	39.21 [36.08, 42.27]
20	40.84 [38.21, 43.71]	41.68 [39.16, 44.39]	42.88 [40.10, 45.71]

TCGA-UT

G	$m = 8$	$m = 16$	$m = 32$
5	61.97 [60.73, 63.09]	62.13 [60.90, 63.23]	62.25 [61.02, 63.34]
10	67.04 [65.78, 68.25]	67.28 [66.01, 68.51]	67.39 [66.10, 68.58]
20	71.76 [70.43, 72.94]	72.02 [70.68, 73.20]	72.08 [70.75, 73.26]

Table 3: Accuracy rises with both factors, but faster down the columns (more patients) than along the rows (more patches per patient), most clearly on TCGA-UT. Entries are patient-macro balanced accuracy (%) with paired-bootstrap 95% intervals; G denotes patients per class and m patches per patient. Scores average splits and draws equally, and intervals condition on the realized training allocations.

at $(20, 8)$. The corresponding TCGA-UT sequence is 62.25%, 67.28%, and 71.76%. Accuracy thus climbs steadily as the same budget is spread over more patients, rather than jumping only between the two extreme cells.

5.2 Breadth and depth contrasts

The contrasts of Equations (7)–(9) condense the grid into three questions: how much does quadrupling depth help, how much does quadrupling breadth help, and what happens when depth is traded for breadth at an unchanged budget? Under nominal support the last answer would be zero. Instead, the equal-budget breadth advantage is 3.71 points on BRACS (95% interval: 2.21 to 5.35) and 9.52 points on TCGA-UT (8.96 to 10.07), with both lower bounds above the one-point practical threshold. How a labelling budget is distributed over patients therefore matters, not only its size.

BRACS

Contrast	Estimate [95% interval]
$\Delta_m(5)$	2.58 [1.87, 3.38]
$\Delta_m(10)$	1.49 [0.46, 2.43]
$\Delta_m(20)$	2.04 [1.13, 2.93]
$\Delta_G(8)$	6.30 [4.78, 7.98]
$\Delta_G(16)$	5.95 [4.39, 7.78]
$\Delta_G(32)$	5.75 [4.34, 7.16]
X	3.71 [2.21, 5.35]

TCGA-UT

Contrast	Estimate [95% interval]
$\Delta_m(5)$	0.28 [0.16, 0.41]
$\Delta_m(10)$	0.34 [0.23, 0.45]
$\Delta_m(20)$	0.32 [0.24, 0.41]
$\Delta_G(8)$	9.80 [9.24, 10.36]
$\Delta_G(16)$	9.88 [9.31, 10.45]
$\Delta_G(32)$	9.84 [9.27, 10.41]
X	9.52 [8.96, 10.07]

Table 4: Quadrupling breadth helps far more than quadrupling depth in both cohorts, and trading depth for breadth at a fixed budget (X) still gains several points. Depth adds about two points on BRACS but less than half a point on TCGA-UT. Contrasts are in percentage points with paired 95% intervals; $\Delta_m(G)$ quadruples depth from 8 to 32 at fixed breadth, and $\Delta_G(m)$ quadruples breadth from 5 to 20 at fixed depth.

The two cohorts part ways on depth (Table 4). On BRACS, additional patches from the same patients keep helping: at twenty patients per class, quadrupling depth improves accuracy by 2.04 points (1.13 to 2.93), so the prespecified saturation criterion is not met. On TCGA-UT, the same contrast is 0.32 points (0.24 to 0.41), and all three depth-gain intervals lie within ± 1 point. Beyond eight patches per participant, TCGA-UT depth is practically saturated.

Breadth gains at fixed depth range from 5.75 to 6.30 points on BRACS and from 9.80 to 9.88 points on TCGA-UT, with every lower interval bound above one point. Because these contrasts also quadruple the patch budget, the equal-budget contrast remains the cleaner evidence that patients matter beyond the number of labelled patches.

5.3 Support surface

The intraclass correlation determines how quickly further patches from one patient lose value, because a patient's m patches count as $m/\mathrm{DE}(m)$ independent observations. With the mean BRACS correlation of 0.130, the design effect rises from 1.91 at depth 8 to 5.02 at depth 32, so a patient's 32 patches still count as roughly six independent observations instead of four. With the mean TCGA-UT correlation of 0.457, the design effect rises from 4.20 to 15.17, and 32 patches count as roughly two observations, barely more than eight patches do. At $G = 20$, class-averaged effective support accordingly rises from 86.94 to 140.26 on BRACS but only from 38.92 to 43.58 on TCGA-UT. The redundancy account thus predicts useful depth on BRACS and almost none on TCGA-UT, which is the pattern of the depth contrasts.

BRACS

Predictor	Slope [95% interval]	Residual SD
$\log n$	2.90 [2.30, 3.54]	1.32
$\log G$	4.33 [3.32, 5.45]	0.98
$\log N_{\text{eff}}$	4.31 [3.36, 5.33]	0.30

TCGA-UT

Predictor	Slope [95% interval]	Residual SD
$\log n$	3.66 [3.46, 3.87]	3.12
$\log G$	7.10 [6.69, 7.50]	0.18
$\log N_{\text{eff}}$	7.07 [6.66, 7.47]	0.25

Table 5: Effective support places the nine BRACS cells closest to a single line, whereas on TCGA-UT patient count alone fits at least as well; nominal support fits worst in both cohorts. Each row fits $A = \alpha + \beta \log(\cdot)$ over the nine pooled cell scores. Slopes are percentage points per natural log unit with paired 95% intervals; residual standard deviations are percentage points with seven residual degrees of freedom and describe in-sample fit, not predictive validation.

A good predictor should place all nine cells on one line, so a smaller residual standard deviation means that the predictor captures more of what makes one allocation better than another. On BRACS, effective support leaves 0.30 points, compared with 0.98 for breadth alone and 1.32 for nominal support (Table 5). The residual breadth coefficients ask the same question from the other side: how much patient count still matters once the support predictor is known. Next to nominal support, breadth retains $\gamma_n = 2.86$ points per log unit (1.74 to 4.07). Next to effective support, it shrinks to $\gamma_e = 0.13$ (-1.60 to 1.85). On BRACS, therefore, accounting for within-patient redundancy absorbs nearly all of the breadth effect; the interval is still too wide to exclude a residual of practical size.

TCGA-UT behaves differently. Breadth alone fits slightly more closely than effective support (residual SD 0.18 versus 0.25 points), and nominal support fits poorly at 3.12 points. After adjustment for effective support, breadth retains $\gamma_e = 4.30$ points per log unit (3.56 to 4.99), or 2.98 points (2.47 to 3.46) per doubling of patients at fixed effective support, with the lower bound above the practical threshold (Table 6). On TCGA-UT, additional patients contribute more than their additional effective observations.

The reciprocal mean correlations place the depth at which redundancy starts to dominate near 7.7 patches on BRACS and 2.2 on TCGA-UT. Both lie at or below the smallest depth, yet BRACS accuracy keeps rising with depth, because its effective support still grows by about

Dataset	Adjustment	γ [95% interval]	Residual SD
BRACS	Nominal	2.86 [1.74, 4.07]	0.29
BRACS	Effective	0.13 [-1.60, 1.85]	0.32
TCGA-UT	Nominal	6.87 [6.46, 7.28]	0.12
TCGA-UT	Effective	4.30 [3.56, 4.99]	0.12

Table 6: Adjusting for redundancy removes the residual breadth effect on BRACS but leaves a substantial one on TCGA-UT. Entries are residual breadth coefficients γ from $A = \alpha + \beta \log(\cdot) + \gamma \log G$, in points per natural log unit with paired 95% intervals; the support slope is refitted jointly, and residual SD uses six residual degrees of freedom. A coefficient of 1.44 corresponds to one point per doubling of breadth.

61% from depth 8 to 32, compared with 12% on TCGA-UT. The transition marks where returns begin to diminish, not where they end.

5.4 Secondary endpoints

Accuracy only asks whether the most probable class is correct. Negative log-likelihood also asks how much probability the model places on the true class, and calibration error asks whether stated confidence matches observed accuracy. Temperature scaling removes any uniform over- or underconfidence, so a difference between allocations that survives it cannot be repaired by one post-hoc scalar. The fitted temperatures show that such a correction is far from trivial: they range from 0.10 to 12.9 on BRACS (median 0.67) and from 0.53 to 1.61 on TCGA-UT (median 0.85), so many unscaled predictions were markedly under- or overconfident.

After scaling, the broad allocation still lowers macro negative log-likelihood at the equal budget, by 0.122 nats on BRACS (95% interval: 0.086 to 0.162) and by 0.389 nats on TCGA-UT (0.372 to 0.408), close to the unscaled differences of 0.113 and 0.463 nats. Calibration error behaves differently. Unscaled, the broad allocation lowers it by 9.39 points on BRACS and 6.59 points on TCGA-UT; after scaling, the reductions shrink to 0.38 points on BRACS (-0.34 to 2.19) and 0.69 points on TCGA-UT (0.34 to 1.13). Scaled calibration error lies between 5.73% and 8.48% across the BRACS cells and between 1.30% and 2.20% across the TCGA-UT cells (Table 7). The raw calibration advantage of breadth is therefore almost entirely a matter of confidence scale, whereas its likelihood advantage is not. Patch-micro balanced accuracy rises by 3.99 points (2.64 to 5.13) and 9.55 points (9.01 to 10.10), close to the corresponding patient-macro gains, so the equal-budget conclusion also does not depend on aggregating patches within patients first.

Depth again matters less than breadth. At twenty patients per class, quadrupling depth lowers scaled negative log-likelihood by 0.054 nats on BRACS and by 0.010 nats on TCGA-UT, a small fraction of the corresponding breadth differences.

Class-specific patient-macro recalls (Table 10) show that the equal-budget advantage is not carried by a single class. On TCGA-UT, all 30 cancer types gain, from 1.44 points for pheochromocytoma and paraganglioma to 19.27 points for uterine carcinosarcoma, and every lower interval bound exceeds zero. On BRACS, six of seven lesion classes gain, led by invasive carcinoma at 9.09 points (6.46 to 11.71); atypical ductal hyperplasia is the sole negative estimate at -2.29 points (-5.54 to 0.87) and its interval contains zero. Each cell's class recalls average to its reported balanced accuracy by construction, so these entries decompose the primary endpoint rather than adding an independent test.

5.5 Robustness

The pooled estimates could hide a single favourable split or a lucky patient draw. Neither is the case. Equal-budget breadth gains are positive in every split, ranging from 3.49 to 3.99 points

BRACS

(G, m)	Scaled NLL (nats)	Scaled ECE (%)	Patch-micro BA (%)
(5, 8)	1.69 [1.64, 1.74]	8.48 [7.19, 12.64]	37.72 [35.77, 39.61]
(5, 16)	1.67 [1.61, 1.72]	8.21 [7.14, 11.72]	39.22 [36.98, 41.40]
(5, 32)	1.65 [1.58, 1.71]	7.86 [6.75, 11.59]	40.73 [38.32, 43.01]
(10, 8)	1.61 [1.55, 1.67]	7.41 [6.33, 10.84]	41.21 [38.94, 43.23]
(10, 16)	1.60 [1.53, 1.67]	8.09 [6.60, 12.07]	41.66 [39.06, 43.90]
(10, 32)	1.57 [1.50, 1.65]	7.44 [6.20, 11.44]	43.05 [40.30, 45.42]
(20, 8)	1.52 [1.46, 1.59]	7.48 [6.34, 10.13]	44.72 [42.14, 46.95]
(20, 16)	1.49 [1.42, 1.56]	6.48 [5.65, 9.81]	46.06 [43.29, 48.50]
(20, 32)	1.47 [1.39, 1.55]	5.73 [5.06, 9.59]	47.43 [44.40, 50.01]
X	-0.12 [-0.16, -0.09]	-0.38 [-2.19, 0.34]	3.99 [2.64, 5.13]

TCGA-UT

(G, m)	Scaled NLL (nats)	Scaled ECE (%)	Patch-micro BA (%)
(5, 8)	1.34 [1.29, 1.39]	2.10 [1.72, 2.93]	62.55 [61.29, 63.63]
(5, 16)	1.33 [1.29, 1.38]	2.20 [1.86, 2.99]	62.72 [61.48, 63.80]
(5, 32)	1.32 [1.28, 1.38]	2.10 [1.78, 2.94]	62.83 [61.58, 63.92]
(10, 8)	1.11 [1.07, 1.16]	1.86 [1.40, 2.73]	67.53 [66.23, 68.71]
(10, 16)	1.10 [1.06, 1.16]	1.78 [1.37, 2.66]	67.77 [66.48, 68.96]
(10, 32)	1.10 [1.06, 1.15]	1.77 [1.38, 2.63]	67.88 [66.59, 69.06]
(20, 8)	0.94 [0.89, 0.99]	1.41 [1.04, 2.25]	72.38 [71.05, 73.53]
(20, 16)	0.93 [0.88, 0.98]	1.33 [1.00, 2.18]	72.64 [71.31, 73.78]
(20, 32)	0.93 [0.88, 0.98]	1.30 [1.00, 2.18]	72.70 [71.37, 73.84]
X	-0.39 [-0.41, -0.37]	-0.69 [-1.13, -0.34]	9.55 [9.01, 10.10]

Table 7: Probabilities improve with breadth even after temperature scaling: scaled negative log-likelihood falls down the columns in both cohorts, whereas scaled calibration error differs little between cells. Entries are per-cell estimates with paired 95% intervals, computed from the stored test outputs of the same fits after one validation temperature per fit. X is the equal-budget contrast $(20, 8)$ minus $(5, 32)$; a negative entry favours the broad allocation for the two probability-quality endpoints and a positive entry favours it for accuracy.

on BRACS and from 8.73 to 10.37 on TCGA-UT (Table 8), and depth gains at twenty patients range from 1.70 to 2.60 points on BRACS and from 0.27 to 0.37 on TCGA-UT. Each split thus reproduces the pooled pattern of useful depth on BRACS and negligible depth on TCGA-UT.

Dataset / split	X	$\Delta_m(20)$	$\Delta_G(8)$	Draw SD range
BRACS / 0	3.65	1.70	6.23	0.33–3.11
BRACS / 1	3.99	2.60	6.75	0.79–2.60
BRACS / 2	3.49	1.82	5.90	1.35–3.58
TCGA-UT / 0	9.46	0.32	9.81	0.34–1.94
TCGA-UT / 1	8.73	0.37	9.09	0.48–1.03
TCGA-UT / 2	10.37	0.27	10.49	0.30–1.48

Table 8: Every split favours breadth at the equal budget and reproduces the cohort difference in depth gains. Contrasts are in points and average five draws per cell. Draw SD is computed within each split and cell with divisor five, and the displayed range spans the nine cells. Splits overlap and are not independent cohort replications.

Draw variation is appreciable, particularly on BRACS. Within-split, within-cell standard deviations span 0.33–3.58 points on BRACS and 0.30–1.94 on TCGA-UT. These are descriptive dispersions over five training draws, not intervals for a fresh training cohort; the paired-bootstrap intervals above describe held-out patient variation conditional on those draws.

Class correlations are reported in Table 9. BRACS ranges from 0.077 for usual ductal hyperplasia to 0.252 for invasive carcinoma; TCGA-UT ranges from 0.314 for lung squamous cell carcinoma to 0.641 for kidney renal clear cell carcinoma. Every TCGA-UT correlation exceeds every BRACS correlation, in line with the cohort difference in depth gains. Within a cohort, however, correlations say little about which classes profit most from breadth. The Spearman rank correlation between class correlation and class-specific equal-budget gain is 0.61 across the seven BRACS classes and -0.31 across the 30 TCGA-UT classes (exploratory $p = 0.15$ and 0.10). On TCGA-UT the sign is even opposite to the redundancy prediction, since classes with more redundant patients gain slightly less from additional patients, which again points to a breadth benefit that the scalar correlation does not capture.

The surface conclusions do not hinge on the estimation error of these correlations. Refitting the augmented effective-support model to the pooled cell scores reproduces the reported coefficients to within 0.03 points. Multiplying every class correlation by a common factor between 0.75 and 1.25, about one to two standard errors on BRACS and more than three on TCGA-UT, keeps the BRACS residual breadth coefficient between -0.58 and 0.82 points per log unit, inside the practical band, and the TCGA-UT coefficient between 2.91 and 5.21, above it. The TCGA-UT residual vanishes only when all correlations are about 1.56 times as large as estimated, a mean of 0.71 instead of 0.46. The nine cells fit that alternative exactly as well, with a residual standard deviation of 0.12 points in both cases, so the grid itself cannot distinguish a larger redundancy from a breadth effect beyond redundancy.

6 Discussion

Spreading a fixed labelling budget over more patients improves patch classification in both cohorts. At 160 patches per class, twenty patients contributing eight patches each outperform five patients contributing 32 patches each by 3.7 points on BRACS and 9.5 points on TCGA-UT, a gain obtained without a single additional label. Because every patient contributes equally, the same patients appear at every depth, and shallow patch sets are nested in deeper ones, the gain cannot be attributed to unequal loss weights or to a change of patients along the depth axis. The grid thus turns the patient-shortage deficit of the preceding experiments into an allocation rule: within the tested range, a labelling budget buys more accuracy when it is spread over patients than when it is concentrated in them.

On BRACS, within-patient redundancy explains most of why patients matter. Effective support orders the nine cells almost perfectly, and once it is known, patient count adds essentially nothing. The moderate correlations also explain why depth keeps helping: further patches from the same patient remain partly informative, only less so than patches from a new patient. A BRACS patient therefore behaves much like a cluster of partially redundant observations, the picture on which the design effect of the taxonomy rests. The interval of the residual coefficient still admits a practically relevant residual, so this account is the most economical description of the grid rather than a proven mechanism.

TCGA-UT departs from this picture. Its patches are highly redundant within participants, depth is practically saturated beyond eight patches, and each doubling of participants still adds about three points more than the implied gain in effective observations. Both properties trace back to how the dataset was built. TCGA-UT is not a tiling of annotated tissue: for each slide, pathologists outlined at least three uniform tumour regions that avoided non-cancerous structures, and ten crops at random angles were taken from every region at each of six field sizes between 128 and 256 μm (Komura et al., 2022). Three regions thus yield 180 patches, which is the median number per training participant in 82 of the 90 class–split combinations. A participant's patches are therefore rotated, rescaled, and partly overlapping views of a few deliberately homogeneous regions. BRACS patches instead grid each region of interest completely into non-overlapping tiles of 64 μm at 40x, without further selection (Brancati et al., 2022), so a patient's patches cover the full extent of its annotated lesions. This difference matches the correlations, every one of which is higher on TCGA-UT than on BRACS, and it explains the saturated depth axis: beyond a few crops, further patches of a TCGA-UT participant mostly repeat tissue that is already represented.

The same construction suggests why the redundancy adjustment leaves a residual on TCGA-UT. Overlapping crops of one region share their tissue in every feature direction, whereas the intraclass correlation measures dependence along a single principal direction that is fitted across all 30 cancer types and need not align with the variation within any one of them. The scalar estimate may therefore understate how redundant these patches are. The grid cannot rule this out, because correlations about 1.5 times as large as estimated would absorb the residual and fit the nine cells exactly as well (Section 5.5). A breadth effect beyond redundancy remains equally compatible with the data. Uniform regions compress the variation within a participant but not between participants, and TCGA-UT pools slides from many submitting institutions whose staining and scanning signatures deep models can recognize (Howard et al., 2021), a source of between-participant variation that the single-institution BRACS cohort lacks. A redundancy measure that is not confined to one direction, such as correlations averaged over the leading principal directions or computed in the subspace that the classifier uses, would separate the two readings without refitting any classifier: if the residual reflects redundancy, the resulting effective support should absorb it.

Temperature scaling shows which part of the probability advantage is merely a matter of confidence level. One temperature makes all predictions of a model more or less confident together, so it corrects systematic over- or underconfidence but cannot shift probability towards the correct class where the model confuses classes. After scaling, broad and deep allocations are about equally well calibrated, so the unscaled calibration advantage of breadth was a difference in confidence level. The likelihood advantage remains, at 0.12 nats on BRACS and 0.39 nats on TCGA-UT, because models trained on more patients place more probability on the correct class, which no rescaling of confidence reproduces. This agrees with the benchmark, where the patient-shortage probability deficit likewise outlasted temperature scaling (Queisler, 2026b). Temperature scaling is worth applying to any allocation, but it does not substitute for patient breadth.

Whichever reading of the TCGA-UT residual holds, the practical recommendation is the same: under a fixed labelling budget, more patients are a better investment than more tissue from the patients already included. In datasets built like TCGA-UT, from repeated crops of a few curated regions per slide, additional depth adds almost nothing beyond eight patches per participant.

7 Limitations

The common eligibility rule restricts training to patients with at least 32 patches of the relevant class. These patients may differ systematically from patients with less tissue or fewer slides. Pairing patients and nesting patches removes changes in eligibility and sampled patient identity from within-draw depth contrasts, but does not establish generalization to excluded patients. Validation and test populations remain unchanged. Depth effects also depend on the deterministic slide-balanced patch ordering used here.

The grid is bounded by the rarest class in each cohort, so its largest allocation stays well below the benchmark's balanced allocations and the absolute accuracies of this experiment are not comparable with those of the earlier reports. The design also fixes equal depth for every patient, which removes contribution inequality but makes the allocations less similar to naturally collected cohorts, where patients differ widely in how much material they supply. Conclusions about the shape of the support surface therefore transfer to naturally unequal cohorts only through the design-effect model.

The redundancy adjustment inherits the approximations of the taxonomy. Its correlation is estimated from a one-dimensional projection of the representation under an exchangeable within-patient correlation model, so dependence that the leading principal direction does not express is absorbed into the residual breadth term. The support surface itself is a descriptive least-squares summary over nine cells; it establishes which predictor orders the observations, not a causal or generative account of the learning problem. All estimates are conditional on frozen Virchow2 features, on multinomial logistic regression, and on the three fixed splits, whose partitions overlap across splits.

Finally, the experiment locates the deficit in the training material but cannot say whether the missing information is absent from the cohort or merely inexpressible in this representation. A residual breadth term is compatible with both, and separating them requires varying the representation rather than the allocation. The probability-quality endpoints inherit their own limits. A single validation temperature corrects only uniform over- or underconfidence, not class-specific miscalibration, and expected calibration error over ten equal-width bins is unstable at these classwise sample sizes, so it is read only alongside the negative log-likelihood.

A Class-level correlations

The class estimates below average the three training-split correlations. Design effects follow Equation (3). They describe dependence along the leading principal direction and do not quantify all between-patient variation in the representation.

BRACS

Class	ICC_c	$\mathrm{DE}_c(8)$	$\mathrm{DE}_c(32)$
Normal	0.109	1.77	4.39
Pathological benign	0.168	2.18	6.22
Usual ductal hyperplasia	0.077	1.54	3.39
Flat epithelial atypia	0.100	1.70	4.11
Atypical ductal hyperplasia	0.084	1.58	3.59
Ductal carcinoma in situ	0.118	1.82	4.65
Invasive carcinoma	0.252	2.76	8.80

TCGA-UT

Class	ICC_c	$\mathrm{DE}_c(8)$	$\mathrm{DE}_c(32)$
Brain Lower Grade Glioma	0.429	4.00	14.29
Breast invasive carcinoma	0.448	4.14	14.88
Glioblastoma multiforme	0.337	3.36	11.44
Lung adenocarcinoma	0.398	3.78	13.32
Lung squamous cell carcinoma	0.314	3.20	10.74
Uterine Corpus Endometrial Carcinoma	0.485	4.40	16.04
Sarcoma	0.493	4.45	16.27
Head and Neck squamous cell carcinoma	0.329	3.30	11.20
Kidney renal clear cell carcinoma	0.641	5.48	20.86
Thyroid carcinoma	0.520	4.64	17.12
Bladder Urothelial Carcinoma	0.508	4.56	16.76
Stomach adenocarcinoma	0.452	4.16	15.00
Skin Cutaneous Melanoma	0.462	4.23	15.31
Prostate adenocarcinoma	0.527	4.69	17.34
Liver hepatocellular carcinoma	0.521	4.64	17.14
Colon adenocarcinoma	0.394	3.76	13.21
Kidney renal papillary cell carcinoma	0.564	4.95	18.49
Cervical squamous cell carcinoma and endocervical adenocarcinoma	0.515	4.61	16.97
Testicular Germ Cell Tumors	0.493	4.45	16.29
Adrenocortical carcinoma	0.500	4.50	16.51
Pancreatic adenocarcinoma	0.328	3.29	11.16
Thymoma	0.358	3.50	12.08
Esophageal carcinoma	0.410	3.87	13.71
Kidney Chromophobe	0.489	4.42	16.16
Ovarian serous cystadenocarcinoma	0.513	4.59	16.91
Uterine Carcinosarcoma	0.378	3.64	12.71
Mesothelioma	0.414	3.90	13.85
Rectum adenocarcinoma	0.530	4.71	17.42
Uveal Melanoma	0.329	3.30	11.19
Pheochromocytoma and Paraganglioma	0.636	5.45	20.72

Table 9: Class-specific intraclass correlations and implied design effects at depths 8 and 32. Larger values indicate stronger projected within-patient dependence.

B Class-specific recalls at the equal budget

The entries below are patient-macro recalls in percent at the two cells of the $n_c = 160$ set, with X their difference. Each column's class mean is the cell's patient-macro balanced accuracy.

BRACS

Class	$(5, 32)$	$(20, 8)$	X
Normal	56.23 [47.65, 63.88]	60.72 [51.91, 68.09]	4.48 [-0.34, 9.46]
Pathological benign	22.63 [19.31, 26.30]	25.45 [20.47, 30.83]	2.82 [-0.87, 6.14]
Usual ductal hyperplasia	22.84 [19.05, 26.79]	26.88 [21.75, 32.00]	4.04 [0.38, 8.44]
Flat epithelial atypia	42.24 [37.33, 48.23]	44.53 [35.95, 53.72]	2.30 [-4.13, 8.07]
Atypical ductal hyperplasia	18.32 [14.04, 23.39]	16.04 [11.87, 21.36]	-2.29 [-5.54, 0.87]
Ductal carcinoma in situ	33.16 [26.41, 40.55]	38.69 [28.86, 50.19]	5.53 [1.09, 11.32]
Invasive carcinoma	64.51 [53.22, 75.19]	73.60 [63.39, 83.08]	9.09 [6.46, 11.71]

TCGA-UT

Class	$(5, 32)$	$(20, 8)$	X
Brain Lower Grade Glioma	64.51 [61.06, 67.85]	77.94 [73.48, 81.65]	13.42 [11.45, 15.36]
Breast invasive carcinoma	64.01 [60.60, 67.23]	71.82 [68.14, 75.32]	7.81 [6.26, 9.42]
Glioblastoma multiforme	67.46 [63.58, 71.38]	73.99 [69.45, 78.15]	6.53 [4.77, 8.31]
Lung adenocarcinoma	48.21 [44.57, 51.71]	60.03 [55.90, 63.62]	11.81 [10.18, 13.47]
Lung squamous cell carcinoma	48.25 [45.28, 51.04]	56.82 [53.10, 60.09]	8.57 [6.33, 10.62]
Uterine Corpus Endometrial Carcinoma	60.25 [55.15, 64.64]	73.32 [67.77, 78.13]	13.07 [10.75, 15.31]
Sarcoma	61.92 [57.91, 65.69]	68.96 [64.04, 73.76]	7.04 [4.89, 9.25]
Head and Neck squamous cell carcinoma	60.72 [55.83, 65.44]	67.10 [61.72, 72.22]	6.38 [4.81, 7.92]
Kidney renal clear cell carcinoma	82.43 [76.30, 87.33]	86.30 [80.26, 90.83]	3.87 [2.56, 5.36]
Thyroid carcinoma	89.04 [85.26, 91.62]	95.05 [91.55, 97.23]	6.02 [4.88, 7.31]
Bladder Urothelial Carcinoma	31.31 [27.92, 34.70]	48.91 [44.15, 53.69]	17.60 [14.25, 21.03]
Stomach adenocarcinoma	36.12 [32.72, 39.62]	53.96 [49.11, 58.73]	17.85 [14.95, 20.86]
Skin Cutaneous Melanoma	58.86 [54.29, 63.18]	66.08 [61.64, 70.38]	7.22 [4.71, 9.88]
Prostate adenocarcinoma	74.27 [70.24, 77.63]	79.31 [74.84, 83.05]	5.04 [3.24, 7.10]
Liver hepatocellular carcinoma	77.89 [72.68, 82.27]	85.34 [79.90, 89.58]	7.46 [5.85, 9.26]
Colon adenocarcinoma	44.77 [41.04, 48.06]	51.55 [47.39, 56.17]	6.78 [3.53, 9.82]
Kidney renal papillary cell carcinoma	76.15 [69.43, 81.73]	81.08 [74.69, 86.04]	4.93 [3.35, 6.97]
Cervical squamous cell carcinoma and endocervical adenocarcinoma	30.61 [26.18, 35.15]	46.63 [40.21, 53.19]	16.03 [12.21, 20.23]
Testicular Germ Cell Tumors	82.61 [74.39, 89.49]	87.13 [79.26, 92.54]	4.52 [2.34, 7.39]
Adrenocortical carcinoma	69.02 [54.87, 80.06]	72.14 [57.73, 83.37]	3.12 [0.22, 5.59]
Pancreatic adenocarcinoma	69.78 [62.13, 76.31]	77.52 [69.67, 83.93]	7.75 [4.42, 11.49]
Thymoma	53.55 [46.90, 59.41]	63.33 [56.50, 69.02]	9.78 [7.31, 12.33]
Esophageal carcinoma	39.23 [34.46, 43.18]	51.94 [46.24, 58.40]	12.71 [7.83, 18.49]
Kidney Chromophobe	71.49 [64.56, 77.72]	85.14 [79.27, 90.07]	13.66 [9.37, 18.69]
Ovarian serous cystadenocarcinoma	72.80 [67.42, 77.67]	84.02 [78.29, 88.59]	11.21 [8.72, 13.67]
Uterine Carcinosarcoma	49.88 [40.38, 59.35]	69.15 [56.99, 79.68]	19.27 [14.70, 24.43]
Mesothelioma	61.84 [47.85, 73.19]	76.52 [62.18, 86.51]	14.68 [9.41, 20.43]
Rectum adenocarcinoma	49.16 [40.49, 56.57]	60.70 [49.07, 69.86]	11.54 [6.08, 16.85]
Uveal Melanoma	78.87 [67.90, 88.45]	87.28 [79.07, 94.58]	8.40 [4.53, 12.93]
Pheochromocytoma and Paraganglioma	92.39 [83.08, 97.33]	93.83 [84.38, 98.65]	1.44 [0.74, 2.40]

Table 10: Class-specific patient-macro recall at the broad allocation $(20, 8)$ and the deep allocation $(5, 32)$, with paired 95% intervals. Positive X favours breadth.

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